OWL SUITS SDR

HMD

Rice University

Apr 3 2025

Introduction

Team

Development Team

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Human Factors & UX Team

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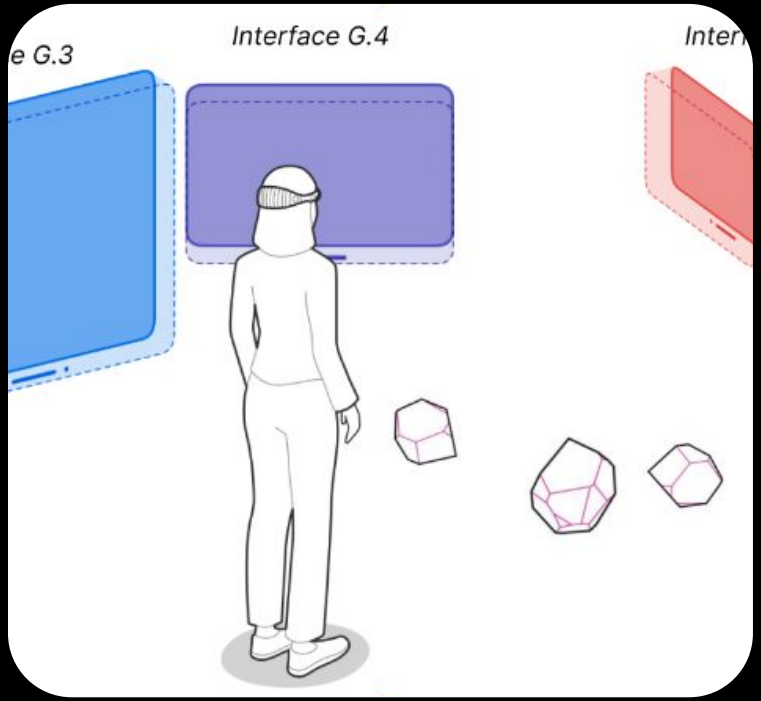
Faculty Advisor: Raudel Avila

- Rooted in human factors (HF) and user-centered design (UCD) principles
- **Just-in-time (JIT) instructions** to minimize cognitive load
- **Visibility of critical data** while maintaining unobstructed view

Human Centered

- **Streamline** routine EVA tasks
- Reduce astronaut **cognitive workload**
- **Minimize potential errors** during mission-critical operations
- **Enhance visibility** under challenging lunar lighting conditions

Outcomes



Key Interface Features

Real-time telemetry visualization

Intuitive navigation and mapping systems

Efficient geological sampling support

Comprehensive caution and warning system

Software Development Plan

Timeline (Dev)

Feb

1. Initial project setup and **TSS integration**
2. HoloLens 2 environment configuration
3. Core architecture **design and prototyping**

Mar

1. Ported **old components** from previous project
2. Implemented basic framework for **ingress procedures**
3. Developed **navigation system** with waypoint tracking

Apr

1. Polish and optimize existing functionality
2. Complete **geological sampling** interface
3. Integration with **RISD rover team** for interoperability

May

1. Final optimizations for low-light conditions
2. On-site testing preparation
3. Deployment package finalization

Timeline (HF)

Feb

1. Created **initial wireframes** and information architecture
2. Established **color palette for lunar lighting conditions**
3. Developed **task flow diagrams** for key operations

Mar

1. Refined visual design language for consistency
2. Created **high-fidelity UI mockups** for development team
3. Designed **interaction patterns for gesture and voice controls**

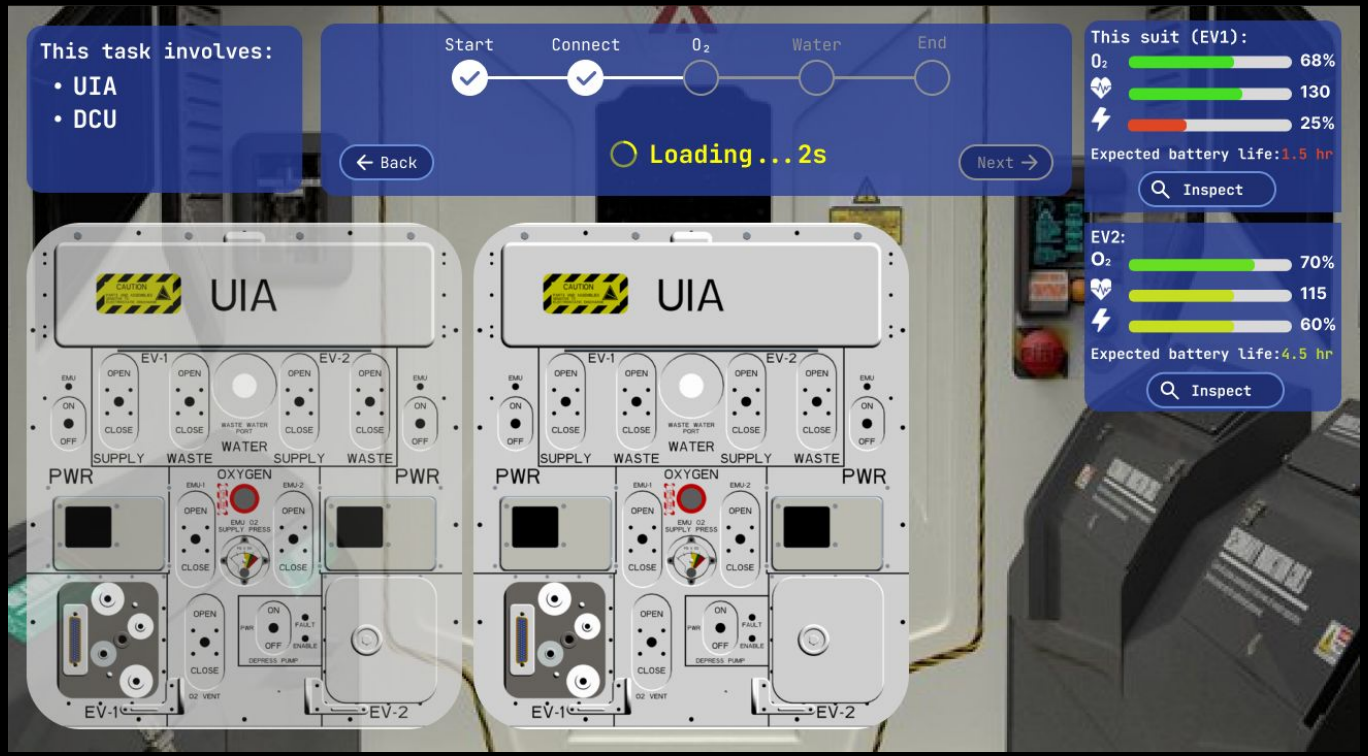
Apr

1. **Human-in-the-loop testing** with subject participants
2. **Usability evaluations** and accessibility refinements
3. Iterative improvements based on test feedback

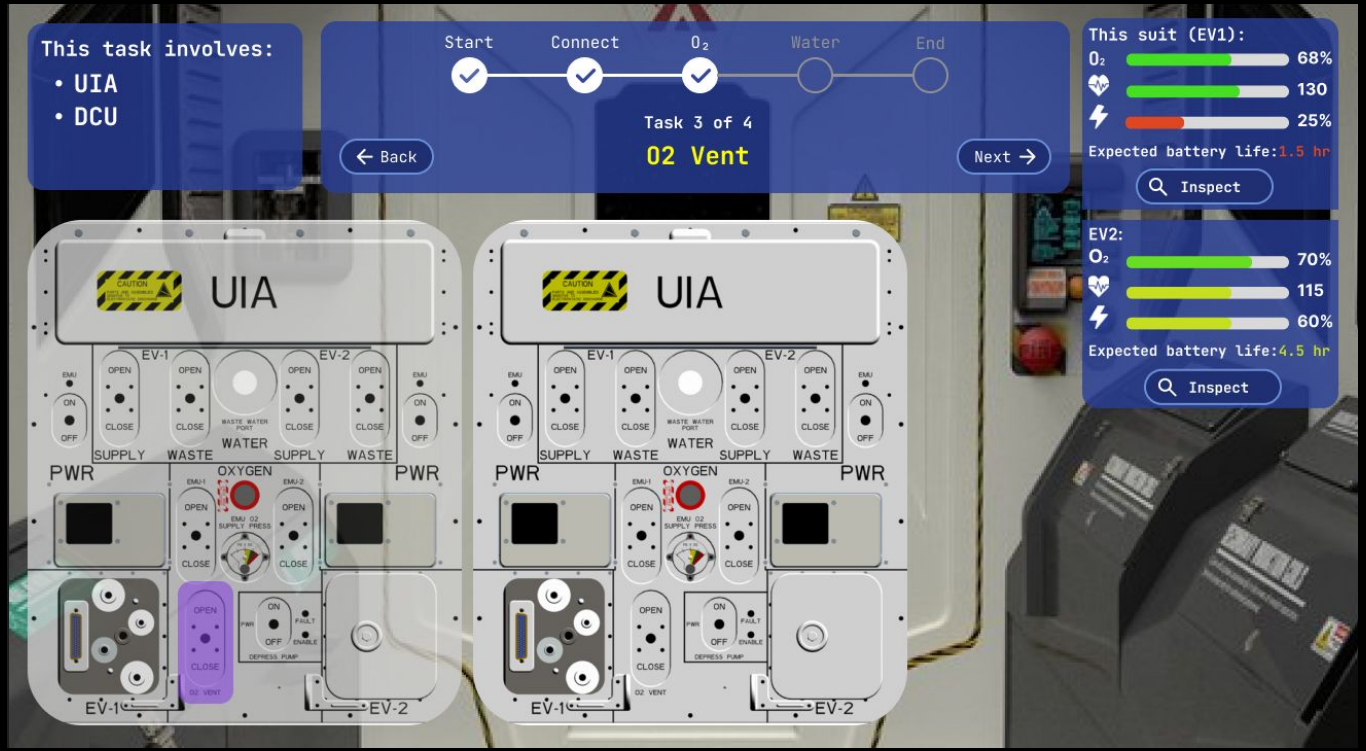
May

1. **Final UI optimizations** based on testing results
2. **Documentation** of design system components
3. Preparation for on-site demonstration

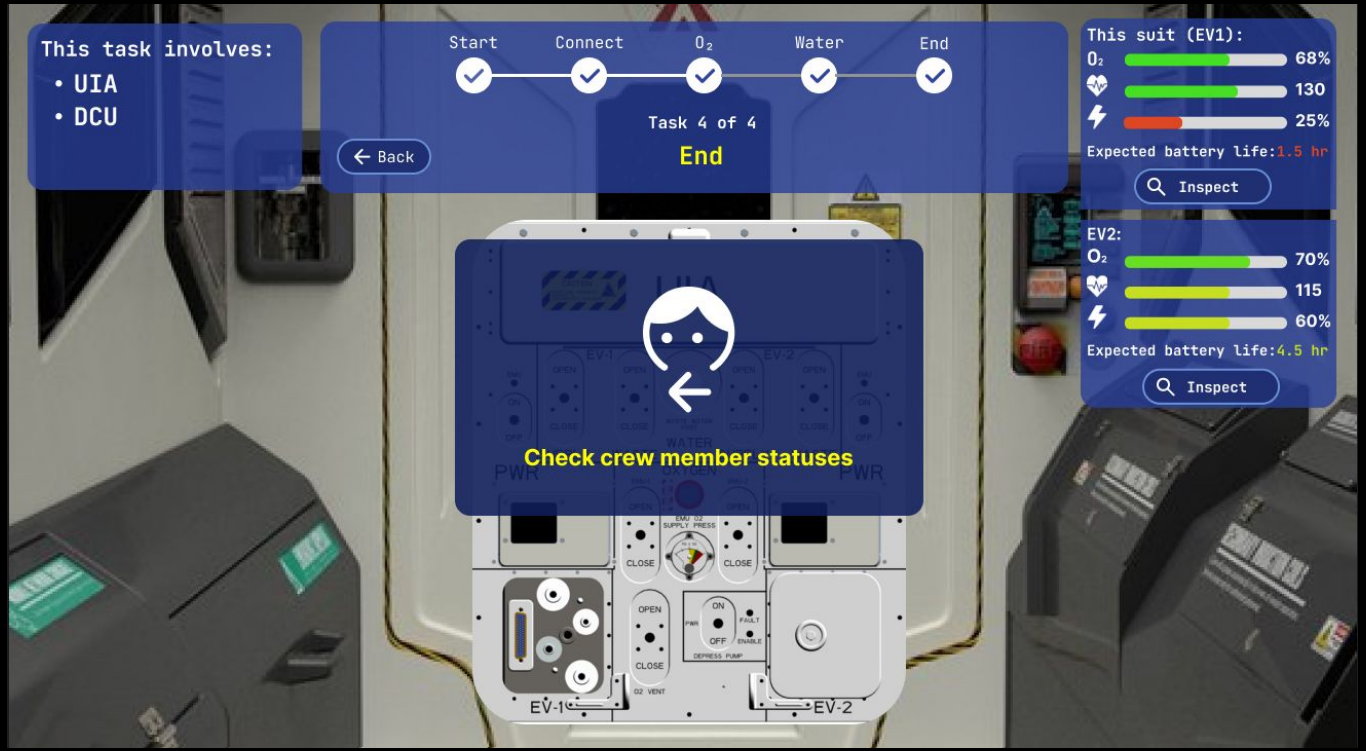
Architecture & Design



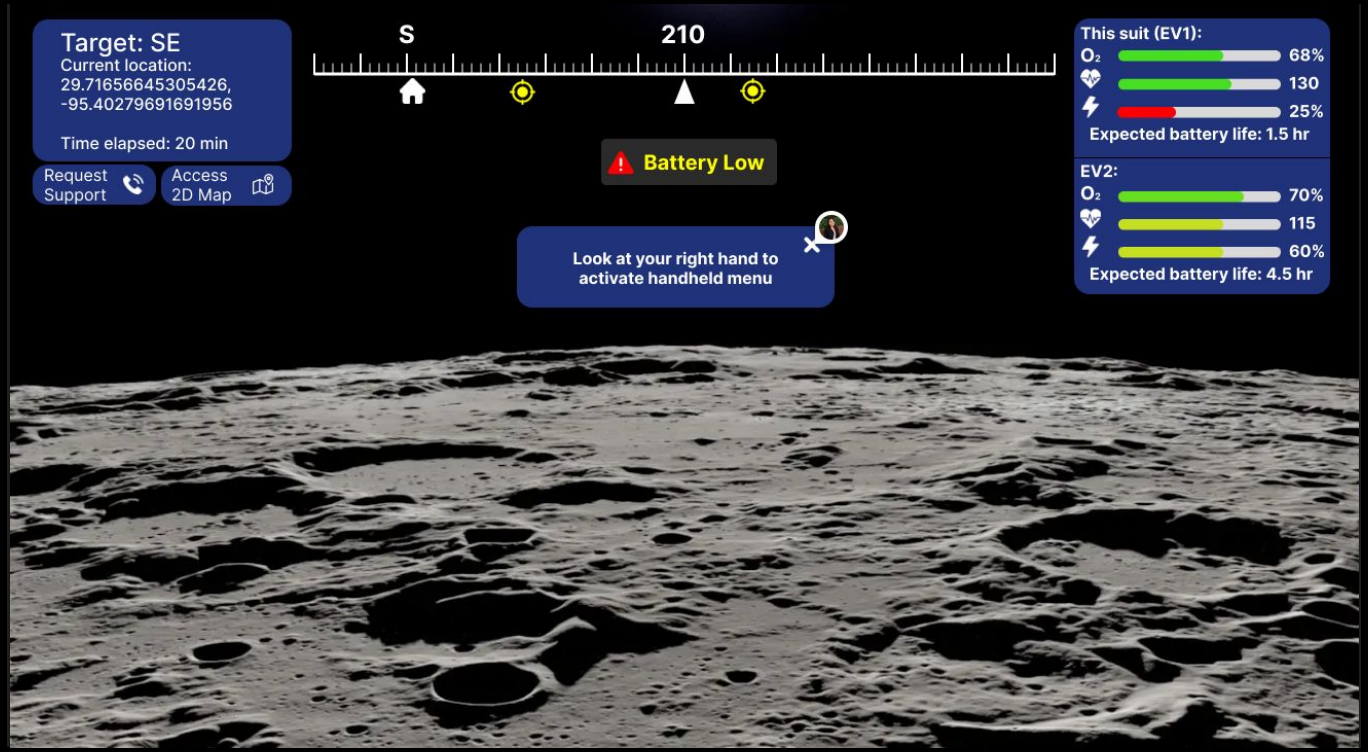
Egress & Ingress



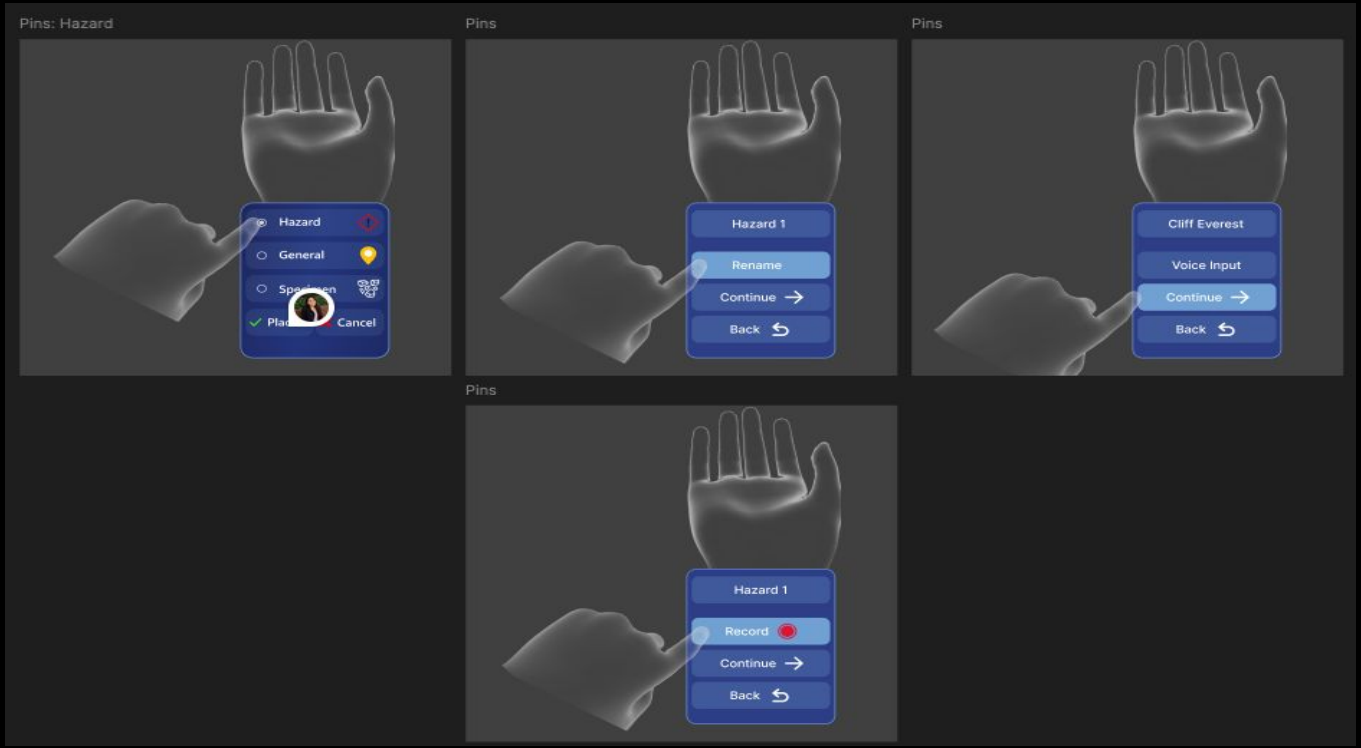
Egress & Ingress



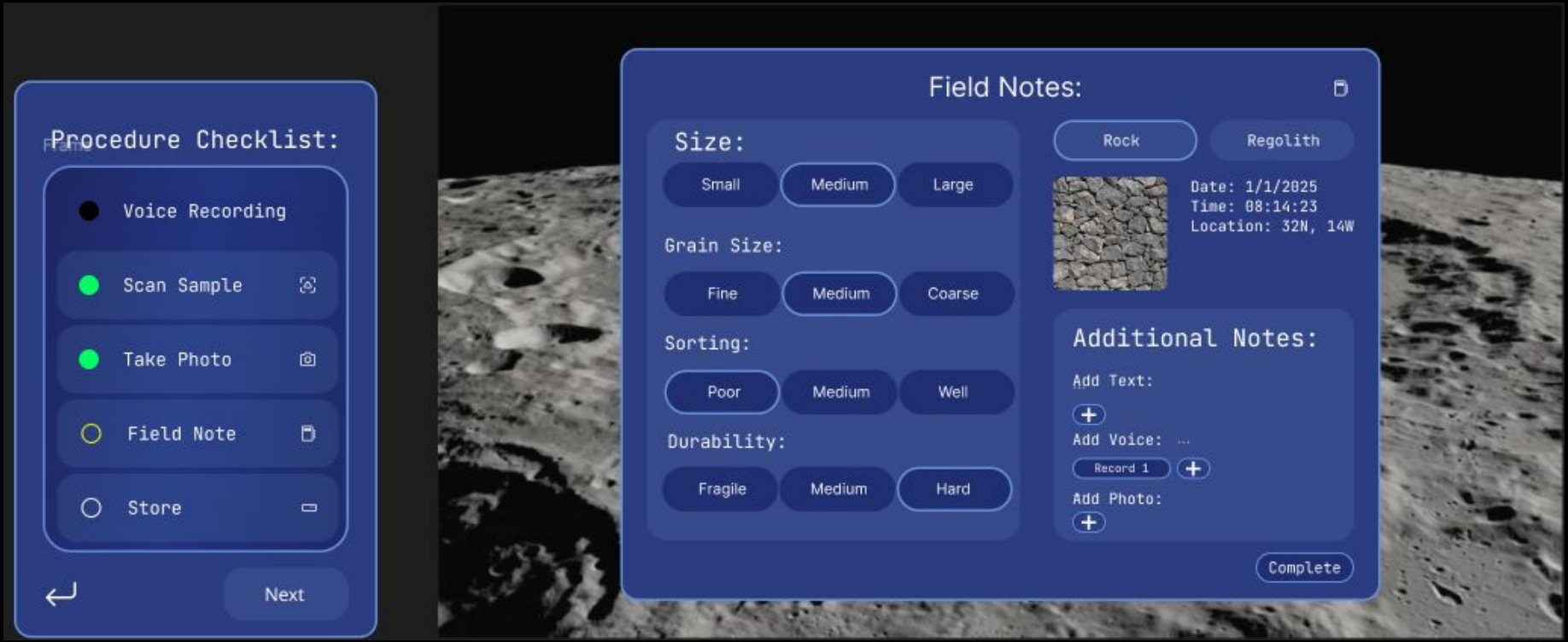
Egress & Ingress


Target: SE
Current location:
29.71656645305426,
-95.40279691691956
Time elapsed: 20 min
Request Support
Access 2D Map
S
210
Battery Low
Look at your right hand to activate handheld menu
This suit (EV1):
O2 68%
130
25%
Expected battery life: 1.5 hr
EV2:
O2 70%
115
60%
Expected battery life: 4.5 hr

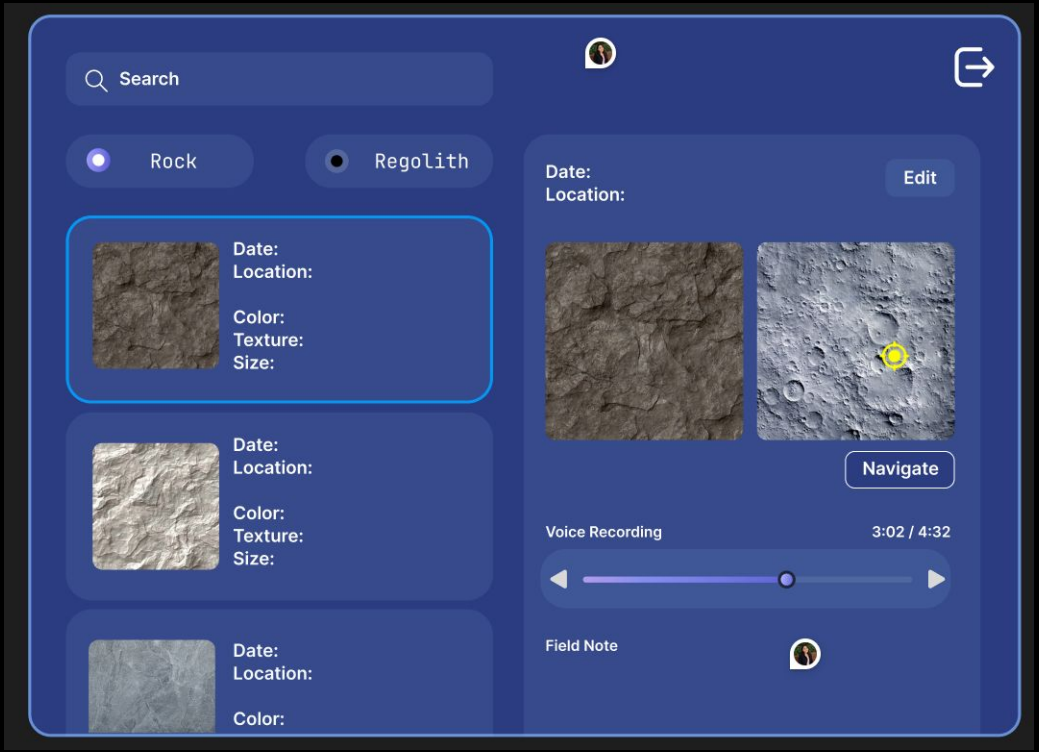
Navigation


Pins: Hazard
Hazard
General
Cancel
Pins
Hazard 1
Rename
Continue
Back
Pins
Cliff Everest
Voice Input
Continue
Back
Pins
Hazard 1
Record
Continue
Back

Navigation



Geological Sampling



Geological Sampling

1. TSS data processing pipeline
2. Interoperability communication layer
3. Task-specific UI components

Components

1. Telemetry data → Processing → Visualization
2. User inputs → Command processing → System actions
3. Asset locations → Mapping system → Navigation UI

Data Flow

- IRB Approval
- Task Analysis
- Cognitive Walkthrough
- User Failure Modes Effects Analysis

Formative Evaluation

Metrics (ISO 9241-11):

- **Efficiency:** task time
- **Effectiveness:** ability to complete the task, number of errors, number of assistance requests, subjective workload (via NASA-TLX)
- **Satisfaction:** subjective usability (via System Usability Scale)

HITL

Participants: Rice undergraduate students will be recruited through the university's online SONA platform

Four Scenarios:

1. Ingress
2. Egress
3. Navigation
4. Geological Sampling

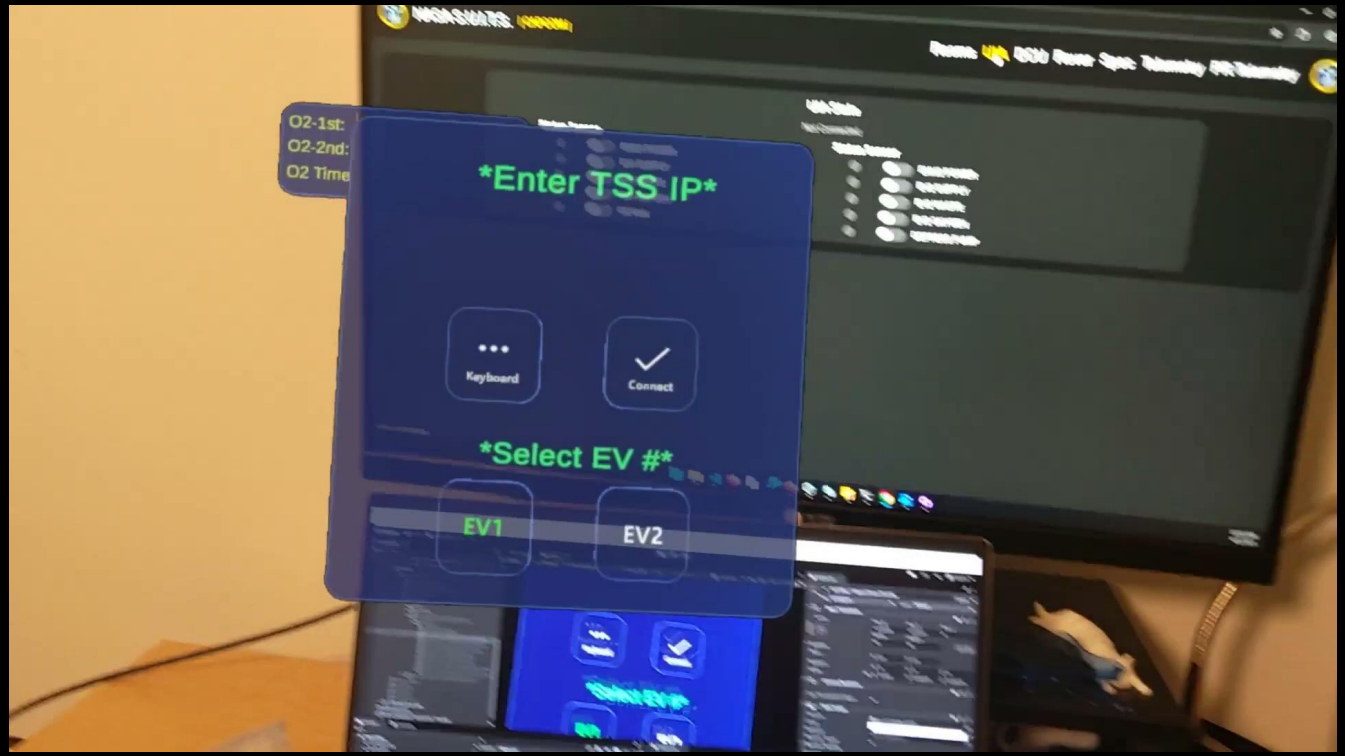
HITL

Expected Results

Since we're testing on an undergraduate sample, many usability issues might be related to unfamiliarity with the tasks astronauts must perform.

However, we expect low to moderate cognitive workload, few errors and assistance requests, and moderate to high usability due to our iterative design process.

HITL



Ingress Demo

Begin HITL testing via Rice SONA platform

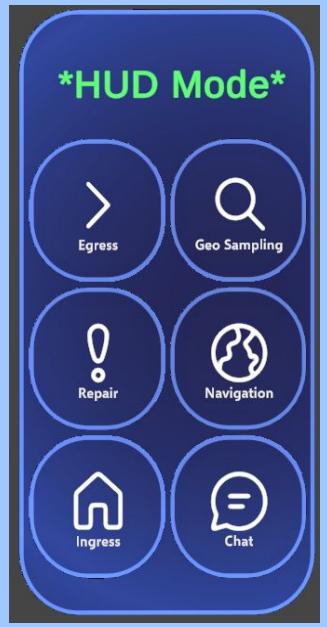
Integrate HMD development with RISD's Rover development

Complete new Geo sampling interface

Integrate new Unity UI with **MRTK3** design language

Optimize UI for low-light conditions.

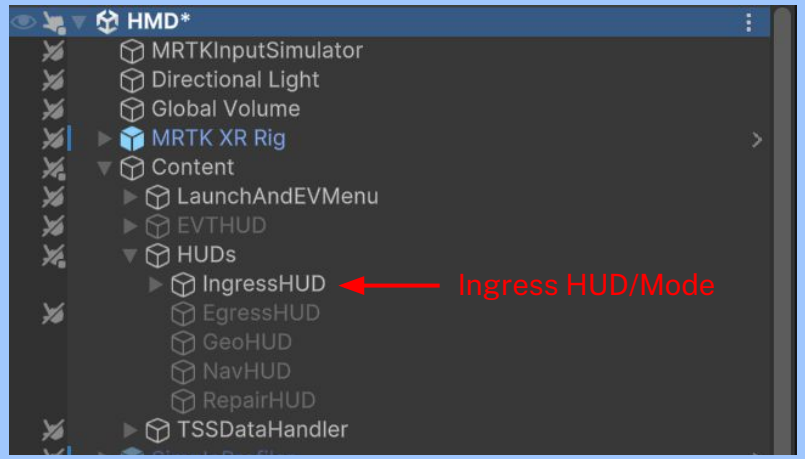
Future Steps



MRTK2 → MRTK3

Background

Interface contained in 1 Unity scene
Modularization achieved via "HUDs" toggled on/off via code.



Current Challenge

Version Control

GitHub

Future Steps + Challenges

Temporary Workaround

Team members create copies of **HMD** scene to develop their components independently. Manually integrate changes into main **HMD** scene.

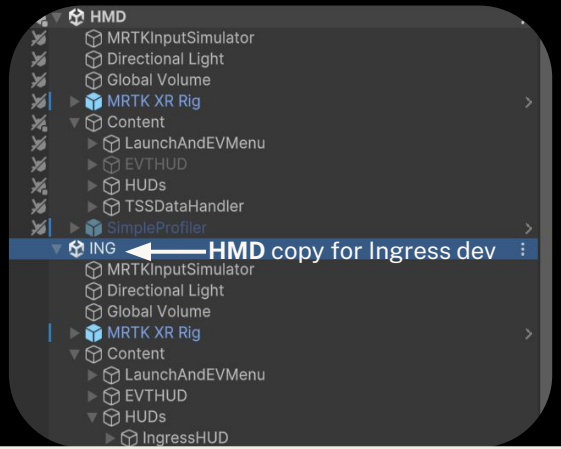
Git Large File Storage (LFS)

Plan to integrate Git LFS, an open-source Git extension designed to handle large files more efficiently.
This will help in tracking changes to substantial assets like Unity scene files.

Game.ci

Plan to integrate Game.ci, which provides Docker-based solutions for Unity projects. This will streamline development environment setup for new team members, in addition to benefits like automated builds and testing.

Solutions



Git Large File Storage

GAME CI

thank you